THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A student heats 100 g of ice and records temperatures. The graph has a flat portion at 0 °C and another at 100 °C. The flat portions represent: 1

(a) cooling **(b)** freezing
(c) melting and boiling **(d)** heating of solid and liquid

2. Which of the following is true about a freely falling stone? 1

(a) it moves in a circle
(b) it is at rest
(c) it is in linear motion and its speed increases
(d) it is in periodic motion

3. If material C is an opaque metal and D is opaque wood, how would you distinguish them? 1

(a) C conducts heat and D does not **(b)** both float
(c) both transparent **(d)** both dissolve

4. If a 20 °C object touches a 60 °C object, which gets warmer? 1

(a) the 60 °C object **(b)** the 20 °C object **(c)** both cool **(d)** neither

5. If 50 mL of water evaporates from a vessel in 5 days, the average loss per day is: 1

(a) 50 mL **(b)** 250 mL **(c)** 5 mL **(d)** 10 mL

6. If the actual length of an object is 8.0 cm and a student measures 8.3 cm, 8.2 cm and 8.4 cm, the error is likely: 1

(a) systematic (consistently higher) **(b)** zero
(c) random only **(d)** due to units

7. Which of the following is a reason to use thermocol for packing glass items? 1

(a) it is transparent **(b)** it is a metal
(c) it is light and cushions shocks **(d)** it is hard

8. A student heats a metal rod and a glass rod at one end for 1 minute. She touches the other ends. She will find: 1

(a) no change **(b)** glass rod hotter **(c)** both same **(d)** metal rod hotter

9. Why does a puddle disappear faster than a lake in the same sun? 1

(a) lake is salty
(b) smaller volume and large surface to depth ratio
(c) puddle is hotter
(d) lake has no water

10. A wheel of circumference 2.5 m makes 480 turns. Distance moved is: 1

(a) 480 m **(b)** 2400 m **(c)** 120 m **(d)** 1200 m

11. An iron cube and a wooden cube of the same size are placed in water. The iron sinks and the wood floats. What does this show? 1

(a) iron is soluble **(b)** iron is heavier for the same size
(c) wood is heavy **(d)** wood is magnetic

12. Which conclusion follows if the temperature of an object is not changing though heat is flowing in and out? 1

(a) object is hot **(b)** no heat flow
(c) heat in equals heat out **(d)** object is cold

13. Which factor does not affect the rate of evaporation? 1

(a) wind **(b)** area exposed
(c) the colour of the container **(d)** temperature

14. Which of the following shows both rotational and linear motion at the same time? 1

(a) a rolling ball **(b)** a pendulum **(c)** a swing **(d)** a fan

15. In an experiment, a mixture of sand, salt and iron filings is given. The first step of separation should be: 1

(a) filtering **(b)** adding water **(c)** using a magnet **(d)** heating

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why do we choose rubber for the handle of an electrician's tools? 2

17. Why do some thermometers use alcohol coloured red instead of mercury, especially at low temperatures? 2

18. How does a glass of cold drink 'sweat' even though the glass has no holes? 2

19. Why is the motion of a swing considered periodic but that of a falling leaf not? 2

20. How can the same material produce different objects? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Why do hill stations often receive more rainfall? 3

22. A drawing is made to a scale of 1 cm for 5 m. A building is 40 m long. What is its length in the drawing? 3

23. Why are bells and utensils often made of alloys like brass? 3

24. What are the possible errors in using a thermometer to read the temperature of a liquid? 3

OR

Describe the effects of climate change on the water cycle (basic). 3

25. A thread wound 25 times round a pencil of uniform thickness measures 8 cm in length of the coil. Find the thickness of the thread. 3

OR

Describe an experiment to show that air is a poor conductor of heat using materials of your choice. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A student says 'My measurement of 12.4 cm is accurate because I measured only once'. Evaluate and suggest improvements. 5

OR

Describe an experiment to show that the same material in different forms (flat sheet and a boat shape) behaves differently in water. 5

27. Compare traditional methods (matka, quilt, mud houses) with modern ones (fridge, thermos, thermocol) for controlling temperature. 5

OR

Analyse: 'Heating curve of ice' with data: 0 min -10 °C; 4 min 0 °C; 14 min 0 °C; 22 min 100 °C; 60 min 100 °C. 5

END OF QUESTION PAPER Total: 50 Marks

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